Ukraine: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- 17 of Ukraine's 28 placed sectors sit in the two low-energy-security corners; against other economies, its Energy Security is lowest in downstream solar, midstream nuclear and downstream wind.[1]
- Openings are scarce: only 1 placed sector sits in the two high-opportunity corners, and its Economic Opportunity is highest in downstream green hydrogen and downstream nuclear.[1]
- The Strategic Index ranks upstream solar 1st among Ukraine's own sectors. When data-based priority is set against policy activity, upstream solar, upstream batteries and midstream solar combine top priority with the most policy, while 6 sectors combine top priority with the least policy.[1]
- The Partnership Strength Index rates Austria highest for Ukraine, as an export market. China, its 1st trade partner, ranks 38th on Partnership Strength.[1,2]

1. Energy security risks and why

17 of Ukraine's 28 placed sectors, those whose scores can both be compared with other economies, sit in the two low-energy-security corners. The Secure corner, which pairs low energy security with low opportunity, holds all 17 of them. Energy Security measures how well a country can rely on what it needs in a sector, from own production and reserves to import needs; higher means more secure.[1]

Against other economies in the same sector, its Energy Security is lowest in downstream solar, midstream nuclear and downstream wind. It scores 0.36, 0.39 and 0.38 there, higher than only 4%, 9% and 18% of the other economies scored in each sector. Each of them sits in the Secure corner.[1]

With no sector in the Upgrade corner, Country Insights takes Ukraine's top Strategic Index sectors as high-priority: upstream solar, downstream solar and upstream batteries.[1]

In downstream solar, the score is pulled down mainly by investment and energy access. Its investment score is 0.00, against a median of 0.40: annual investment scores 0.00, and growth in investment is not reported by the source, which the index counts as zero. Its solar consumption was 0.557 GJ per person in 2025, higher than only 18% of the other economies on that measure.[1,3]

In midstream nuclear, trade is the only category behind the score, and it pulls the score down: its trade deficit in these goods was 0.1% of GDP in 2023. On that measure it scores higher than only 1% of the other economies. For context, Sweden supplied 91.1% of its imports of these goods in 2024.[1,2,4]

2. Economic opportunities and why

Openings are scarce: only 1 of Ukraine's placed sectors, downstream green hydrogen, sits in the two high-opportunity corners. Economic Opportunity measures how well placed a country is to create value in a sector, from exports and market share to investment, costs, demand and related capabilities.[1]

Against other economies in the same sector, its Economic Opportunity is highest in downstream green hydrogen (Grow corner), downstream nuclear (Watch) and downstream oil (Secure). Its scores there are higher than 64%, 44% and 40% of the other economies scored.[1]

In downstream green hydrogen, consumption pushes the score up: on use per person today it scores higher than 83% of the other economies, and on expected growth in use per person higher than 81%.[1]

In downstream nuclear, investment pushes the score up: on annual investment it scores higher than 92% of the other economies. Cost competitiveness pulls it down: on input costs for labour and capital it scores higher than only 5%.[1]

In downstream oil, production pushes the score up: power generation from oil was 0.842 TWh in 2025, and on growth in production it scores higher than 91% of the other economies.[1,3]

The paper's framework associates the Grow corner with export promotion, and the Watch corner, where 10 of its sectors sit, with creating value elsewhere in the chain.[1]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by opportunity, security need, policy activity, stage and low-emission value. Among Ukraine's own sectors, it ranks upstream solar 1st (0.64), downstream solar 2nd and upstream batteries 3rd. The scores of downstream solar and upstream batteries are almost tied, 0.006 apart, and upstream batteries is almost tied with midstream solar (4th).[1]

The rank of downstream solar reflects its policy term of 0.200, the full weight of that term.[1]

Ukraine's allied pick in the network design is upstream batteries (0.62), almost tied with the runner-up, midstream solar (0.002 apart). The pick is Ukraine's first upstream or midstream sector on the Strategic Index after a resource-base screen. The upstream sector ranked above it, upstream solar, is screened out because no resource base is recorded for it.[5]

Country Insights splits the sectors into thirds by data-based priority (the Strategic Index without its policy term) and again by policy activity. On that comparison, upstream solar, upstream batteries and midstream solar combine top priority with the most policy. By contrast, upstream wind, upstream electric grid, upstream green hydrogen, midstream nuclear, midstream electric grid and upstream electric vehicles combine top priority with the least policy. At the other end, downstream oil and downstream coal combine bottom priority with the most policy.[1]

The Climate Action Tracker does not rate Ukraine, so its policy basis is International Energy Agency Policies and Measures records only. Ukraine has 85 such records, 42 of which name a technology. For context, 15 records name downstream solar specifically, against 0 for upstream wind.[1,6]

4. Partnership priorities and why

The Partnership Strength Index scores partners in energy supply chains as reliable suppliers (Friendshore), export markets (Partnership Opportunity) and development partners (Development Potential). It rates

Austria highest for Ukraine (0.80), as an export market, strongest in midstream electric vehicles, midstream wind, midstream solar, downstream nuclear and midstream electric grid.[1]

Germany (2nd, also 0.80), the Czech Republic (3rd) and Poland (4th) follow, each scoring best as a reliable supplier. Georgia ranks 5th, scoring best as a development partner, with a Development Potential of 1.00.[1]

Today's trade matches part of this ranking. On average across its energy-goods baskets, Poland is Ukraine's 2nd trade partner, the Czech Republic its 3rd and Germany its 5th. Austria is only its 13th and Georgia its 28th.[1,2]

China is its 1st trade partner, with 27.4% of its imports on average, but ranks 38th on Partnership Strength, a strong tie today that the index ranks lower. In UN votes they sit in opposite blocs, while the partners named above vote in the same bloc as Ukraine.[1,2,7]

As a partner to the coalition, Ukraine ranks 60th of 192 economies. In the allied network design it is a member with no production share in any of the 22 upstream and midstream stages.[1,5]

5. Data notes and caveats

Values are as of 12 October 2026; Country Insights has more.

- Scores compare Ukraine with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- The Climate Action Tracker does not rate Ukraine, so its policy activity rests on International Energy Agency Policies and Measures records alone; its policy levels compare only with economies on the same basis.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.

Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
2. United Nations, UN Comtrade Database, HS6 bilateral trade (all reporters), 2020 and 2024; reconciled by the authors (pulled 2026-10-07 to 2026-10-09); attribution; https://comtradeplus.un.org/.
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5. Allied network design (the paper's model design of coalition production shares).
6. International Energy Agency, Policies and Measures Database (export of 2026-09-30); attribution; https://www.iea.org/policies.
7. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.